

M2- Chi-Square Test of Independence

Part 1

Independent variable:Race(White/Black/Other)

Dependent variable:Demand(R feels that family make too many demands on R)

H₀: There is no relationship between Race and Demand

H₁: There is a relationship between Race and Demand

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	31.322 ^a	8	<.001
Likelihood Ratio	28.935	8	<.001
Linear-by-Linear Association	18.131	1	<.001
N of Valid Cases	1169		

a. 3 cells (20.0%) have expected count less than 5. The minimum expected count is 2.65.

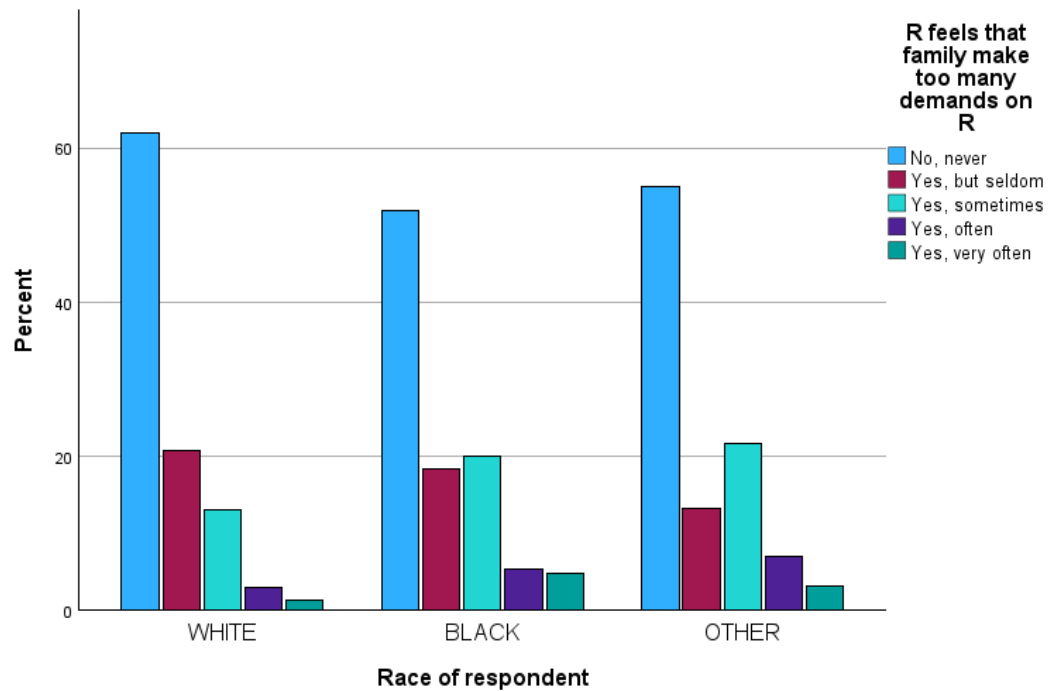
Chi-square Test:The result was statistically significant, $\chi^2=21.322$, $DF=8$, $P<0.01$. Since the p-value is less than 0.05, we rejected H₀. That means the race and demand are significantly associated.

Assumptions:The variables are categorical, and the observations are independent. However, the output shows that 3 cells, or 20.0%, have expected counts less than 5, with a minimum expected count of 2.65. This slightly violates the expected count assumption, so the result should be interpreted with some caution.

R feels that family make too many demands on R * Race of respondent Crosstabulation						
			Race of respondent			
			WHITE	BLACK	OTHER	Total
R feels that family make too many demands on R	No, never	Count	526 ^a	99 ^b	71 ^{a, b}	696
		Expected Count	505.5	113.7	76.8	696.0
		% within Race of respondent	62.0%	51.8%	55.0%	59.5%
	Yes, but seldom	Count	176 ^a	35 ^a	17 ^a	228
		Expected Count	165.6	37.3	25.2	228.0
		% within Race of respondent	20.7%	18.3%	13.2%	19.5%
	Yes, sometimes	Count	111 ^a	38 ^b	28 ^b	177
		Expected Count	128.5	28.9	19.5	177.0
		% within Race of respondent	13.1%	19.9%	21.7%	15.1%
	Yes, often	Count	25 ^a	10 ^a	9 ^a	44
		Expected Count	32.0	7.2	4.9	44.0
		% within Race of respondent	2.9%	5.2%	7.0%	3.8%
	Yes, very often	Count	11 ^a	9 ^b	4 ^{a, b}	24
		Expected Count	17.4	3.9	2.6	24.0
		% within Race of respondent	1.3%	4.7%	3.1%	2.1%
Total		Count	849	191	129	1169
		Expected Count	849.0	191.0	129.0	1169.0
		% within Race of respondent	100.0%	100.0%	100.0%	100.0%

Each subscript letter denotes a subset of Race of respondent categories whose column proportions do not differ significantly from each other at the .05 level.

Post-hoc test: For “No, never,” White respondents are significantly different from Black respondents. For “Yes, sometimes,” White respondents are significantly different from Black and Other respondents. Black and Other respondents do not significantly differ from each other in that category. This suggests that White respondents are more likely to say “No, never,” while Black and Other respondents are more likely to report that family makes too many demands at least sometimes. Others we failed to find a difference.



Both the Chi-Square test and the clustered bar chart indicate that:

There is a significant relationship between race and perceived family demands

White respondents are less likely to report high family demands

Black and Other respondents are more likely to report experiencing family demands at least sometimes

Add a third variable-sex:

R feels that family make too many demands on R * Race of respondent * Respondents sex Crosstabulation

Respondents sex				Race of respondent			Total
				WHITE	BLACK	OTHER	
MALE	R feels that family make too many demands on R	No, never	Count	271 ^a	48 ^a	29 ^a	348
			Expected Count	259.9	56.0	32.0	348.0
			% within Race of respondent	64.2%	52.7%	55.8%	61.6%
		Yes, but seldom	Count	97 ^a	15 ^a	7 ^a	119
			Expected Count	88.9	19.2	11.0	119.0
			% within Race of respondent	23.0%	16.5%	13.5%	21.1%
		Yes, sometimes	Count	43 ^a	18 ^b	10 ^{a, b}	71
			Expected Count	53.0	11.4	6.5	71.0
			% within Race of respondent	10.2%	19.8%	19.2%	12.6%
		Yes, often	Count	8 ^a	7 ^b	3 ^{a, b}	18
			Expected Count	13.4	2.9	1.7	18.0
			% within Race of respondent	1.9%	7.7%	5.8%	3.2%
		Yes, very often	Count	3 ^a	3 ^{a, b}	3 ^b	9
			Expected Count	6.7	1.4	.8	9.0
			% within Race of respondent	0.7%	3.3%	5.8%	1.6%
		Total	Count	422	91	52	565
			Expected Count	422.0	91.0	52.0	565.0
			% within Race of respondent	100.0%	100.0%	100.0%	100.0%
FEMALE	R feels that family make too many demands on R	No, never	Count	255 ^a	51 ^a	42 ^a	348
			Expected Count	246.0	57.6	44.4	348.0
			% within Race of respondent	59.7%	51.0%	54.5%	57.6%
		Yes, but seldom	Count	79 ^a	20 ^a	10 ^a	109
			Expected Count	77.1	18.0	13.9	109.0
			% within Race of respondent	18.5%	20.0%	13.0%	18.0%
		Yes, sometimes	Count	68 ^a	20 ^a	18 ^a	106
			Expected Count	74.9	17.5	13.5	106.0
			% within Race of respondent	15.9%	20.0%	23.4%	17.5%
		Yes, often	Count	17 ^a	3 ^a	6 ^a	26
			Expected Count	18.4	4.3	3.3	26.0
			% within Race of respondent	4.0%	3.0%	7.8%	4.3%
		Yes, very often	Count	8 ^a	6 ^a	1 ^a	15
			Expected Count	10.6	2.5	1.9	15.0
			% within Race of respondent	1.9%	6.0%	1.3%	2.5%
		Total	Count	427	100	77	604
			Expected Count	427.0	100.0	77.0	604.0
			% within Race of respondent	100.0%	100.0%	100.0%	100.0%
Total	R feels that family make too many demands on R	No, never	Count	526 ^a	99 ^b	71 ^{a, b}	696
			Expected Count	505.5	113.7	76.8	696.0
			% within Race of respondent	62.0%	51.8%	55.0%	59.5%
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Each subscript letter denotes a subset of Race of respondent categories whose column proportions do not differ significantly from each other at the .05 level.

Chi-Square Tests

Respondents sex		Value	df	Asymptotic Significance (2-sided)
MALE	Pearson Chi-Square	30.999 ^b	8	<.001
	Likelihood Ratio	26.968	8	<.001
	Linear-by-Linear Association	17.177	1	<.001
	N of Valid Cases	565		
FEMALE	Pearson Chi-Square	13.769 ^c	8	.088
	Likelihood Ratio	12.124	8	.146
	Linear-by-Linear Association	3.937	1	.047
	N of Valid Cases	604		
Total	Pearson Chi-Square	31.322 ^a	8	<.001
	Likelihood Ratio	28.935	8	<.001
	Linear-by-Linear Association	18.131	1	<.001
	N of Valid Cases	1169		

a. 3 cells (20.0%) have expected count less than 5. The minimum expected count is 2.65.

b. 4 cells (26.7%) have expected count less than 5. The minimum expected count is .83.

c. 4 cells (26.7%) have expected count less than 5. The minimum expected count is 1.91.

Part 2:

The purpose of the exercise.

The purpose of this exercise is to examine whether there is a relationship between the race of respondents and their perception of how often their family makes too many demands on them.

How to calculate the statistical technique.

The Chi-Square test of independence is calculated by comparing the observed frequencies in each category with the expected frequencies assuming the variables are independent. The formula is $\chi^2 = \sum((O - E)^2 / E)$, where O represents observed counts and E represents expected counts. The resulting Chi-Square statistic is then used to determine a p-value based on the degrees of freedom.

Interpret the findings.

The results of the Chi-Square test indicate a statistically significant relationship between race and perceived family demands, $\chi^2(8) = 31.322$, $p < .001$. This suggests that the

distribution of responses differs across racial groups. Specifically, White respondents are more likely to report "No, never," while Black and Other respondents are more likely to report experiencing family demands at least sometimes. The clustered bar chart supports these findings by visually showing differences in response patterns across racial groups. Post-hoc tests further indicate that some of these differences are statistically significant between specific groups. Although some expected cell counts were below 5, the large sample size helps support the reliability of the results. **Overall, the findings suggest that race is associated with how individuals perceive family demands.**